

Hyperbolic Flavor Geometry: Mixing, CP Violation, and Fermion Masses from Arithmetic Hyperbolic 3-Manifolds

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Abstract

We present a unified geometric framework in which the Standard Model flavor parameters—CKM and PMNS mixing matrices, the leptonic CP-violating phase, and the charged fermion mass ratios—are encoded in the arithmetic of two compact hyperbolic 3-manifolds: $M_{\text{PMNS}} = m003(-2, 3)$ and $M_{\text{CKM}} = m006(-5, 2)$, both with $H_1 = \mathbb{Z}/5$ and volume below 2.1. The PMNS mixing matrix arises from the Borel factor of the holonomy representation of M_{PMNS} (Frobenius fitness 0.005087 to PDG 2024—the global minimum of the construction). The leptonic CP-violating phase is derived parameter-freely: $\delta_{\text{HFG}} = \pi + \varphi(\text{aaB}) + \varphi(\text{baa}) = 195.91^\circ$ (0.55% from PDG). The trace fields $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary) and $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$ (real) explain the CP asymmetry between sectors geometrically. All charged fermion mass ratios are approximated by norms in $\mathbb{Z}[\omega]$ (leptons: τ error 0.006%) and $\mathbb{Z}[\sqrt{17}]$ (quarks: b error 0.012%, t error 0.0065%), with permutation null-test $p < 0.002$. The framework is falsifiable by Hyper-Kamiokande, DUNE, and CMB-S4.

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1 Introduction

The flavor parameters of the Standard Model—the CKM [10] and PMNS [11] mixing matrices, CP-violating phases, and fermion mass ratios—have no derivation from first principles. We show they are consistent with encoding in two compact arithmetic hyperbolic 3-manifolds of minimal volume with $H_1 = \mathbb{Z}/5$. The construction has four independently testable layers: (i) mixing matrices from Iwasawa decomposition of holonomy; (ii) leptonic CP phase from twist angles, parameter-free; (iii) CP asymmetry from the imaginary/real trace field dichotomy; (iv) fermion mass ratios as norms in the rings of integers of the trace fields.

A direct test shows that primitive geodesic lengths do *not* statistically encode the fermion mass lattice (enrichment factor 0.40, i.e. depletion). The arithmetic encoding operates at a deeper level through the ring-of-integers norm forms.

2 The Two Manifolds

Manifold	Census	Name	Volume	H_1	CS	Sym.
M_{PMNS}	1	$m003(-2, 3)$	0.9814	$\mathbb{Z}/5$	1/4	$\mathbb{Z}_2^2$
M_{CKM}	43	$m006(-5, 2)$	2.0289	$\mathbb{Z}/5$	-0.1141	$\mathbb{Z}_2^2$

2.1 Selection of M_{PMNS}

Among all closed orientable hyperbolic 3-manifolds with $H_1 = \mathbb{Z}/5$ in the first 500 census entries, M_{PMNS} is the unique element satisfying: (a) minimal volume (0.9814, smallest known [4]); (b) Lucas-pure covering tower: all new torsion primes through degree 9 are prime Lucas numbers $\{2, 3, 11\}$, and 13 never appears; (c) a geodesic of length $\frac{3}{2} \log \varphi = 0.72157$ (second shortest, to 0.03%)—the smearing parameter that minimises the CKM construction.

2.2 Trace fields

Proposition 1. $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ and $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$.

Verification. For M_{PMNS} : $\mathbb{Q}(\sqrt{-3})$ is established for the cusped ancestor $m003$ in [3]. For M_{CKM} : $\text{tr}(\rho(aa)) = 3 - \sqrt{17}$, satisfying $x^2 - 6x - 8 = 0$ with discriminant $68 = 4 \cdot 17$, verified at 150-bit precision. Rings of integers: $\mathbb{Z}[\omega]$ (norm $a^2 - ab + b^2$) and $\mathbb{Z}[\sqrt{17}]$ (norm $|a^2 - 17b^2|$). $\square$

3 Mixing Matrices

The axis $\hat{n}(\gamma) \in S^2$ of $\gamma \in \pi_1(M)$ is extracted from $\log \rho(\gamma)$ via Pauli decomposition; pairwise angles define a Gaussian overlap matrix; QR orthogonalization gives a unitary matrix.

CKM. Triple $\{\text{abbAbbaB}, \text{Ab}, \text{B}\}$ in $\pi_1(M_{\text{CKM}})$: $\mathcal{F} = 0.009078$ ($\sigma = 0.420$), 45% improvement over canonical. Jarlskog $J = 0$ (homological class collision).

PMNS. Triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ in $\pi_1(M_{\text{PMNS}})$ with Borel (lower-triangular) overlap: $\mathcal{F} = 0.005087$ —global minimum of the Borel construction; $p < 10^{-4}$ vs 50,000 Haar-random unitary matrices.

4 Geometric CP Phase

Theorem 2 (CP phase, zero free parameters).

$$\delta_{\text{HFG}} = \pi + \varphi(\text{aaB}) + \varphi(\text{baa}) = 195.91^\circ,$$

where $\varphi(\gamma) = \text{Im} \log \lambda(\gamma)$. PDG 2024: 197.0° ; difference 1.09° (0.55%).

Verification. $\varphi(\text{aaB}) = -176.73^\circ$, $\varphi(\text{baa}) = -167.36^\circ$ from 150-bit polished holonomy. Sign fixed by $\det[\hat{n}(\text{aa}), \hat{n}(\text{aaB}), \hat{n}(\text{baa})] = +0.079 > 0$. $\square$

Theorem 3 (CP asymmetry from trace fields). K_{PMNS} is imaginary $\Rightarrow$ loxodromic holonomy $\Rightarrow$ large leptonic CP. K_{CKM} is real $\Rightarrow$ holonomy traces in $\mathbb{R} \Rightarrow$ geometric CP suppression ($J_{\text{CKM}} = 0$).

5 Arithmetic Mass Encoding

Theorem 4 (Arithmetic mass encoding—computational). All charged fermion mass ratios m_f/m_e are well approximated by norms in the rings of integers of the trace fields (Table 1).

Table 1: Best norm witnesses for fermion mass ratios.				
Fermion	Ring	m_f/m_e	Norm	Error
μ	$\mathbb{Z}[\omega]$	206.77	$N(-16 - 12\omega) = 208$	0.59%
τ	$\mathbb{Z}[\omega]$	3477.22	$N(-68 - 37\omega) = 3477$	0.006%
d	$\mathbb{Z}[\sqrt{17}]$	9.14	$ 3^2 - 0 = 9$	1.5%
s	$\mathbb{Z}[\sqrt{17}]$	182.78	$ 14^2 - 17 = 179$	2.1%
c	$\mathbb{Z}[\sqrt{17}]$	2495.11	$ 841 - 3332 = 2491$	0.16%
b	$\mathbb{Z}[\sqrt{17}]$	8180.04	$ 49 - 8228 = 8181$	0.012%
t	$\mathbb{Z}[\sqrt{17}]$	338082	$ 19321 - 357425 = 338104$	0.0065%

Remark 5. The Lucas primes $L_{11} = 199 = N(-15 - 13\omega)$ and $L_{17} = 3571 = N(-69 - 34\omega)$ are also Eisenstein norms (exact), with errors 3.8% and 2.7%; they satisfy $\equiv 1 \pmod{3}$ (split in K_{PMNS}) and $\equiv 3 \pmod{4}$ (not Gaussian norms), arithmetically locking them to K_{PMNS} . The witnesses in the table are the nearest integer norms within search radius $R = 150$; for the muon, $N = 208$ is closer to m_μ/m_e than the Lucas prime $L_{11} = 199$ (0.59% vs 3.8%). The Lucas prime is included in the remark because of its additional arithmetic significance (prime, split, exact Eisenstein norm), not because of superior accuracy.

5.1 Statistical validation

A permutation null test ($N = 500$ shuffles of mass assignments) gives $p < 0.002$ for $\mathbb{Z}[\sqrt{17}]$ on mean log-error (observed -2.605 , null mean -2.098) and on fraction within 2% (observed 0.75, null mean 0.62). Comparison against 10 random real quadratic discriminants shows $d = 37$ achieves marginally better mean log-error (-2.618 vs -2.605); the identification $d = 17$ rests on the independent arithmetic $\text{tr}(\rho(aa)) = 3 - \sqrt{17}$, not on error minimization alone.

6 Covering Tower and Lucas-Purity

Observation 6. Through degree 9, every new torsion prime in finite covers of M_{PMNS} is a prime Lucas number from $\{2, 3, 11\}$; the prime 13 never appears.

Conjecture 7 (Lucas-purity). All torsion primes in the covering tower of M_{PMNS} are prime Lucas numbers. Proof would require connecting torsion growth to Hecke eigenvalues of the quaternion algebra of discriminant 3 via Jacquet-Langlands.

7 Predictions and Open Problems

Falsifiable predictions. (1) $\delta_{\text{PMNS}} = 195.91^\circ$, testable by Hyper-Kamiokande and DUNE ($\sim 5^\circ$ precision). (2) The prime 13 never appears in any finite cover of M_{PMNS} . (3) Neutrino masses (speculative, no direct geometric support): $\sum m_\nu \approx 71$ meV, testable by CMB-S4.

Open problems. (1) Derive m_e from geometric invariants (volume, CS= 1/4, systole). (2) Explain the Jarlskog scale ≈ 0.007 ; proximity to $\alpha_{\text{em}} \approx 1/137$ may or may not be exact. (3) Compute Bianchi modular form Hecke eigenvalues over $\mathbb{Q}(\sqrt{-3})$ (requires MAGMA); test whether they are Lucas numbers. (4) Extend covering tower to degree 12 and beyond. (5) Dirac eigenvalue computation (requires specialized hyperbolic PDE solver). (6) Gauge structure: embed $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ in the construction.

8 Conclusions

We have shown that two arithmetic hyperbolic 3-manifolds encode the CKM and PMNS mixing matrices, the leptonic CP phase (zero free parameters, 0.55% error), and all charged fermion mass ratios (permutation $p < 0.002$). The CP asymmetry between sectors is explained by the imaginary/real trace field dichotomy. The framework is a geometric phenomenology analogous to the Balmer formula: the pattern is established; the dynamical foundation awaits discovery.

Data Availability

Scripts and data: <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

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References

- [1] S. Navas et al., Phys. Rev. D **110**, 030001 (2024).
- [2] M. Culler et al., <http://snappy.computop.org>.
- [3] T. Chinburg, E. Friedman, K. N. Jones, A. W. Reid, Ann. Scuola Norm. Sup. Pisa **30**, 1 (2001).
- [4] D. Gabai, R. Meyerhoff, P. Milley, J. Amer. Math. Soc. **22**, 1157 (2009).
- [5] M. L. Gentry, SSRN:6583550 (2026).
- [6] M. L. Gentry, SSRN:6583549 (2026).
- [7] M. L. Gentry, SSRN:6583553 (2026).
- [8] M. L. Gentry, SSRN:6754501 (2026).
- [9] M. L. Gentry, SSRN:6761981 (2026).
- [10] M. Kobayashi, T. Maskawa, Prog. Theor. Phys. **49**, 652 (1973).
- [11] Z. Maki, M. Nakagawa, S. Sakata, Prog. Theor. Phys. **28**, 870 (1962).